Panels in the coset construction for a C^+-group

Human Compression project

Abstract

We distinguish a rank restriction missing from the literal statement of Corollary 4.9 of Leemans–Tranchida from the unresolved assertion in higher rank. We also give an exact criterion for geometry and firmness in rank four, using products of the four maximal parabolics. This reduces a possible finite counterexample to a small certificate without an automorphism computation. The criterion is checked on four previously acquired markings; none is a counterexample in rank four.

1 The rank restriction

We use the final published edition of [1], printed pages 493–494 and 497. Put $\alpha_0 = 1$ and

$$P_J = \langle \alpha_i^{-1}\alpha_j : i, j \in J \rangle.$$

The C^+ intersection condition is $P_J \cap P_K = P_{J\cap K}$ for $|J|, |K| \geq 2$. Construction 3.2 takes $H_i = P_{I\setminus\{i\}}$ and forms the incidence system of right cosets $H_i g$, with incidence defined by nonempty intersection. A geometry means that every flag extends to a flag containing one object of each type. Connectedness is a separate condition.

Proposition 1. *For every nontrivial cyclic group $G = \langle a \rangle$, the rank-two marking $R = \{a\}$ satisfies the stated C^+ condition and is independent. Its associated coset system is a geometry, but is not firm.*

Proof. Here $I = \{0, 1\}$, $\alpha_0 = 1$, and $\alpha_1 = a$. The only subset of I with at least two elements is I itself, so the required intersection equality is $G \cap G = G$. Independence says $a \notin \langle \varnothing \rangle = 1$, which holds. Both maximal parabolics are trivial, since their defining index sets are singletons. Each type therefore has one object for each $g \in G$. The type-zero object labelled g is incident to the type-one object labelled h exactly when $g = h$. The geometry consists of disjoint edges. Every flag extends to an edge, but every one-object panel has exactly one completion. This is the failure of firmness. □

No lower rank bound is stated in the displayed C^+ definition, Construction 3.2 or Corollary 4.9. The proof of Lemma 4.8, however, chooses an index k distinct from two prescribed indices i, j. Consequently this example identifies a missing rank restriction in the literal wording of both Lemma 4.8 and Corollary 4.9: replacing one face of a matching edge gives a nonincident pair, not an adjacent chamber. Section 2.1, printed page 492, imposes neither connectedness nor a lower rank bound on a geometry, and its Proposition 2.1 permits every positive number of types. The later definitions do not add a global restriction. The counterexample does not decide the intended statement for rank at least three. In rank three the desired conclusion follows from pair transitivity and the canonical adjacent chambers, as in the existing project proof of Proposition 4.11. For rank at least four, the remaining problem is to supply a neighbour for a panel outside the orbit of a canonical panel. The accepted counterexample to the principal residual-connectedness theorem is thin, and does not solve this problem.

2 A criterion in rank four

The next reduction applies to arbitrary subgroups of a finite group; the C^+ condition is needed only for the last description of the stabilizer. In particular, no transitivity on triple flags is assumed.

Lemma 2. *Let G be finite, let $I = \{0,1,2,3\}$, and let $H_i \le G$. For each missing type d, choose an ordering (a,b,c) of $I \setminus \{d\}$ and put*

$$X_c = H_cH_a \cap H_cH_b, \qquad Y_d(x) = H_dH_a \cap H_dH_b \cap H_dH_cx \quad (x \in X_c). \tag{1}$$

The coset system is a geometry if and only if $Y_d(x) \neq \varnothing$ for all four choices of d and all $x \in X_c$. It is a firm geometry if and only if

$$|Y_d(x)| \ge 2|H_d| \quad \text{for all these } d, x. \tag{2}$$

It suffices in either test to take one representative for each orbit of $K = H_a \cap H_b$ on the right cosets H_cx contained in X_c. For a rank-four C^+ marking, $K = P_{\{c,d\}} = \langle \alpha_c^{-1}\alpha_d \rangle$ is cyclic.

Proof. Every incident pair can be taken to H_a, H_b by right multiplication: choose an element z in the intersection of its two cosets and multiply both cosets by z^{-1}. Every pair, every object and the empty flag therefore extend to a canonical chamber after the same normalization. The only possible obstruction to being a geometry is a triple flag with no completion at the fourth type.

After normalizing the types a, b of such a triple, its last object is H_cx. It intersects both H_a and H_b precisely when $x \in X_c$. A type-d coset H_dy completes the triple precisely when y belongs to each of the three sets defining $Y_d(x)$. Each such set, and hence their intersection, is a union of right cosets of H_d. The number of completions is therefore exactly $|Y_d(x)|/|H_d|$. This proves the geometry criterion and (2). Once all panels have at least two completions, every flag of corank one has a residue with at least two objects, which is precisely firmness.

Changing x to hx with $h \in H_c$ leaves $Y_d(x)$ unchanged. Right multiplication by $k \in K$ fixes H_a, H_b and sends the panel with third object H_cx to the panel with third object H_cxk. It bijects their completions; equivalently $Y_d(xk) = Y_d(x)k$. Thus one representative from each stated orbit suffices. Finally the intersection condition gives $H_a \cap H_b = P_{I\setminus\{a,b\}}$. The subgroup for a two-element index set is generated by either one of the mutually inverse differences, giving the cyclic description. □

3 Finite checks and remaining scope

The accompanying program checks the rank-two example for $G = C_2$ by enumerating all flags: there are four objects, two chambers, and four panels, each with one completion. This check is separate from the uniform written proof for all nontrivial cyclic groups.

For rank four the program reuses the four stored $\mathrm{PSL}(2,7)$ markings from the earlier diagnostic. It checks the multiplication table against determinant-one matrices modulo their central signs, regenerates all sixteen index subgroups, and verifies their 256 intersection equalities. It checks (1) against literal coset intersections for every normalized panel, then partitions the panels into the stated stabilizer orbits. No marking is searched for anew. Each of the four geometries has five completions at every panel. The existing projective-line description gives the same conclusion: four typed copies of an eight-point set are incident by inequality.

For a future fixed finite group, a proposed counterexample must pass the nondegeneracy and full intersection checks, have no zero entry in (1), and have at least one entry equal to

$|H_d|$. A singleton panel without the first of these two completion requirements would not refute Corollary 4.9: the coset system might fail to be a geometry. This packet does not produce a counterexample or a general proof in rank at least four, and it gives no whole-original, conjecture, formalization or compression credit.

The C^+ construction and canonical neighbour formula are due to the authors of [1] and their cited predecessors. The previously written chamber-component argument and the four stored markings are reused. The present additions are the explicit rank boundary and the panel-counting reduction; no claim of historical novelty for the elementary coset calculation is made.

References

[1] D. Leemans and P. Tranchida, *On residual connectedness in chiral geometries*, Algebraic Combinatorics **4** (2021), 491–499, doi:10.5802/alco.162.